Predicting the Site Of Origin of Idiopathic Ventricular Arrhythmias using a Machine Learning Approach

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I. Introduction

Ventricular arrhythmias (VAs) are abnormal heart rhythms originating in the ventricles, ranging from premature ventricular contractions to ventricular tachycardia. They can cause significant symptoms such as palpitations, dizziness, syncope, and in severe cases increase the risk of sudden cardiac death.

Idiopathic ventricular arrhythmias (IVAs) are a subgroup of VAs occurring in patients without structural heart disease. They represent approximately 10–15% of ventricular arrhythmias and, although often benign, can lead to a high arrhythmic burden and impaired ventricular function when frequent [1], [2].

Catheter ablation is the main treatment for symptomatic IVAs, and its success strongly depends on accurate identification of the site of origin (SOO). Precise SOO localization improves procedural efficiency, reduces risks, and increases ablation success rates. However, current localization methods based on ECG interpretation and invasive mapping remain challenging, especially across anatomically adjacent regions [3], [4].

Most IVAs originate from the right or left ventricular outflow tracts (RVOT and LVOT), although additional clinically relevant sites exist, including the RVOT septum, RVOT free wall, RCC-LCC commissure, LVOT summit, and LVOT subvalvular regions. These different origins produce distinct ECG morphologies, which can be exploited for non-invasive prediction [1], [4].

Machine learning methods offer the ability to learn complex patterns from ECG data and improve SOO localization. In this context, this study focuses on patients from Teknon Medical Center and investigates machine learning approaches for SOO prediction using ECG-derived signals and clinical data [4], [5].

Objectives

1. Binary classification: distinguish between RVOT and LVOT origins.
2. Multi-class classification: RVOT origins, Commissure (LCC-RCC) & LVOT (summit-subvalvular)
3. Fine-grained classification: perform detailed SOO prediction at a more specific anatomical level.

II. Methods

A. Available Data

Four different data sources were used in this project. The first dataset was collected at Teknon Medical Center in Barcelona and includes 181 patients with idiopathic ventricular arrhythmias. For each patient, 12-lead ECG recordings (4-beat segments) and associated clinical variables were available. The second dataset, from Hospital Clínic de Barcelona, consists of 31 patients with 12-lead ECG segments focused on QRS complexes and annotated ventricular arrhythmia cases. The third dataset is an open-source Chinese database containing 334 patients with 12-lead QRS-centered ECG recordings, used to improve population diversity and model generalization [6]. Finally, a synthetic dataset of 2,469 simulated patients was included, consisting of 12-lead QRS ECG signals generated to augment training data and address class imbalance [7].

Abbreviation	Dataset
Teknon	Real clinical dataset from Teknon Hospital
CARTO	Clinical dataset (CARTO system)
Sims	Simulated dataset
China	External clinical dataset from Teknon

TABLE I: Definition of datasets used in the study.

The overall distribution of the datasets is illustrated in Appendix A.A Figure 10 and Appendix A.B Figure 11.

B. Clinical Variables Preprocessing

Clinical information included sex, age, hypertension (HTA), premature ventricular contraction (PVC) transition, height, weight, body mass index (BMI), diabetes mellitus (DM), dyslipidemia (DLP), smoking status, chronic obstructive pulmonary disease (COPD), sleep apnea, and a clinical score. A missingness analysis is provided in Appendix B.A Figure 12. Due to the limited size of the Teknon cohort, no patients were excluded for missing clinical data. Categorical variables (sex, HTA, DM, DLP, smoking status, COPD, and sleep apnea) were converted into binary features for model compatibility.

Age was available for 178 patients, with a mean of 56.65 ± 15.6 years (range 15–88). Missing values were imputed using the cohort mean. Height and weight were analyzed using boxplots and the interquartile range (IQR) method, defining outliers as values outside $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$. No height outliers were detected. Height was missing in 28 patients and imputed using sex-specific medians. Sex-stratified analysis showed mean heights of 175.29 ± 8.43 cm for males and 163.13 ± 6.62 cm for females. Weight was also missing for 28 patients, with four outliers (114, 117, 120, 122 kg) retained as biologically plausible. Male patients had a mean weight of 85.67 ± 14.02 kg, and female patients 67.31 ± 9.33 kg. Missing values were imputed using sex-specific medians, as this approach is more robust to skewed distributions and extreme values. BMI was computed for all patients using imputed values:

$$BMI = \frac{\text{Weight (kg)}}{\text{Height (m)}^2}$$

PVC transition was included as a key predictor of site of origin, as it reflects ventricular activation patterns used clinically to differentiate RVOT and LVOT origins. For patients with missing values, PVC transition was derived from ECG recordings by measuring R and S wave amplitudes in each precordial lead. The transition lead was defined as the first lead where the R/S ratio was ≥ 1, indicating R-wave dominance.

C. ECG Segmentation

The CARTO, Sims and China datasets provided the QRS waves for a single cardiac beat per recording, whereas the Teknon dataset contained the four complete beats, with the last being the arrhythmic event. Consequently, the Teknon signals required additional segmentation and cropping, which was applied to all 12 leads.

The segmentation of ECG waves was performed using an automated pipeline based on the Jimenez's SAK library and deep learning models [8]. First, the original signals were resampled to 250 Hz using a Fast Fourier Transform (FFT)-based resampling approach. Subsequently, robust amplitude normalization was applied, followed by notch filtering to attenuate power-line interference in the 50-60 Hz frequency range.

The P, QRS and T waves were then classified using the Jimenez's ensemble of five CNN models and SAK. This process yielded the onset, offset, and peak locations associated with each wave, as illustrated in Figure 1. Finally, R-peaks were localized by identifying the absolute maximum amplitude within the detected QRS interval, and were aligned to match second 2. Since the processed data was intended for machine learning applications, which generally require inputs of consistent dimensions, a fixed temporal window was selected. QRS segments shorter than the target length were padded accordingly to ensure dimensional homogeneity across the dataset. Only the segmented QRS complexes were retained for all subsequent analyses and model development in this study, while P and T wave information was discarded.

D. Experimental Protocol

All models, including both deep learning (DL) and classical machine learning (ML) approaches, are evaluated under a unified protocol using exclusively data from Hospital Teknon. Model development is performed using 5-fold cross-validation with validation on the Teknon dataset. For the deep learning

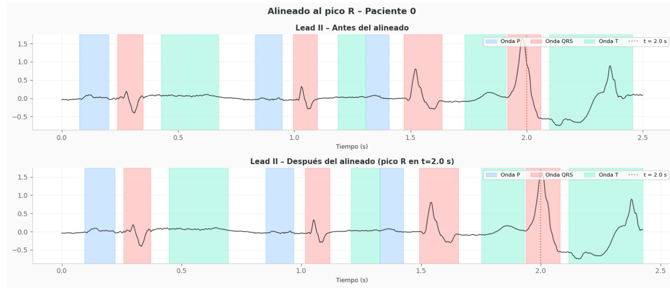

Fig. 1: QRS Segmentation of patient 0 Lead II, before and after peak alignment

and machine learning models, each fold is trained for 20 epochs. An independent hold-out test set, corresponding to the last 20% of the dataset, is reserved and used only for final evaluation.

E. Deep Learning (DL)

A convolutional neural network (CNN) was used for processing 12-lead ECG signals. The architecture is based on the model proposed by Saglietto et al. [9], designed to preserve inter-lead spatial relationships for ECG classification (An schematic representation is provided in Appendix A.C, Figure 11). In this study, the CNN is applied to ECG segments centered on the QRS complex. The network consists of three 1D convolutional blocks, each composed of a convolutional layer, batch normalization, ReLU activation, and max-pooling. The number of filters is progressively increased across layers to capture higher-level morphological and temporal features. Additional pooling operations were included to accommodate the reduced signal length and improve feature compression.

1) Convolutional Neural Network with Multilayer Perceptron: To incorporate clinical information, a second branch based on a multilayer perceptron was added to process eight clinical variables. This branch consists of two fully connected hidden layers, enabling the learning of nonlinear relationships between patient-level features. For the final prediction, the feature representation extracted by the convolutional neural network is combined with the multilayer perceptron output using a late fusion strategy. Both embeddings are concatenated and passed through two fully connected layers, reducing the dimensionality from 16 to 2 for final classification. The entire model was trained using cross-entropy loss as the objective function, with optimization performed using the Adam optimizer.

F. Classical Machine Learning (ML)

Several classical machine learning models were used as baselines. Logistic Regression provides a simple linear classifier. Support Vector Machines capture non-linear relationships using kernel functions. Random Forest models complex interactions through ensemble decision trees, while Gradient Boosting improves performance by sequentially correcting errors. Together, these methods offer a range from interpretable linear models to powerful non-linear ensembles for comparison with deep learning approaches.

G. Difference between DL and Classical ML

The CNN preserves the local spatial and temporal structure of the ECG by learning directly from raw QRS signal segments. Through convolutional filters, it is able to capture morphological patterns and hierarchical representations that reflect the underlying cardiac activity. In contrast, classical machine learning models treat the QRS segment as a flat feature vector, without explicitly exploiting local dependencies or temporal relationships between consecutive samples. As a result, they rely on predefined feature representations rather than learning directly from the raw signal structure.

H. Feature Engineering

For each QRS complex and ECG lead, 73 features were extracted to describe both morphological and statistical properties of the signal. Among them, PVC transition information was included to encode the direction of ventricular depolarization, which is relevant for distinguishing at least RVOT and LVOT origins. To calculate the PVC transition, the R/S ratio is computed across the precordial leads, and the transition is defined as the first lead in which the R/S ratio becomes greater than 1. A summary of extracted features and PVC calculation method is provided in Appendix C.B.1 (Table VI) and Appendix C.B.2 (Equation 1, 2).

I. Evaluation Metrics

The performance of the models was evaluated using the macro F1-score. This metric balances precision and recall, providing a robust measure that penalizes both false positives and false negatives, especially in imbalanced datasets. The macro F1-score assigns equal importance to all classes, ensuring that minority classes contribute equally to the final evaluation. [10]

J. Data Assessment

A data assessment was conducted through multiple deep learning and machine learning experiments. All seven possible dataset combinations including the Teknon cohort (See table II), were evaluated in order to identify the optimal training configuration.

Configuration	Dataset Combination
Config 0	Teknon
Config 1	Sims + Teknon
Config 2	China + Teknon
Config 3	CARTO + Teknon
Config 4	China + Sims + Teknon
Config 5	CARTO + Sims + Teknon
Config 6	CARTO + China + Teknon
Config 7	All datasets (CARTO + China + Sims + Teknon)

TABLE II: Dataset configurations used in the study.

III. Experiments

A schematic summary of the experiments is provided in Appendix C.A Table VII for clarity.

A. Binary Classification - LVOT/RVOT

Raw Data

The raw ECG-based experiments were focused on the binary classification task, aiming to distinguish between right and left ventricular outflow tract (RVOT vs LVOT) origins.

1) Experiment 1: CNN - dataset assessment: A dataset assessment was performed for the CNN using raw ECG data from the datasets.

2) Experiment 2: CNN + MLP - all datasets: A convolutional neural network with a multilayer perceptron (CNN + MLP) architecture was trained using all available datasets combined (Configuration 7). In this setting, clinical variables from the Teknon cohort were incorporated through the MLP, allowing the integration of Teknon's patient-level information into the model.

3) Experiment 3: Classical ML - dataset assessment: Four classical machine learning algorithms were evaluated using the same dataset assessment framework.

Feature-Based Approach

4) Experiment 4: ML - Data Assessment: Feature extraction was applied to all datasets, representing each QRS complex with 73 features. A data assessment was then performed to determine the most optimal dataset combination for model performance using these feature-based representations.

5) Experiment 5: ML on Best Data Combination with Clinical Variables from Teknon: The same machine learning approach was applied using the best-performing dataset combination, this time including clinical variables from the Teknon cohort, in order to evaluate whether clinical information improves prediction performance. Since the additional datasets did not contain these clinical variables, missing values were imputed to ensure a consistent feature representation across all datasets. For the best-performing model, hyperparameter optimization was subsequently performed using a randomized search with 5-fold cross-validation. A total of 30 hyperparameter combinations were randomly sampled from a predefined search space. The best configuration was selected based on the highest cross-validation Macro F1-score and was finally evaluated on the unseen Teknon hold-out test set.

6) Experiment 6: ML on the Teknon Dataset with Clinical Variables: To further investigate the impact of clinical information, the same classical machine learning methodology was applied using only the Teknon dataset and its associated clinical variables. This experiment was designed to assess whether clinical features provide additional predictive value when evaluated independently of the external datasets. For the best-performing model, the same hyperparameter optimization as in the previous experiment was performed.

B. Three-Class Classification

For the three-class classification task, a feature-based approach using classical machine learning models was applied.

1) Experiment 7: ML Data Assessment and Model Optimization: For the three-class classification task, a dataset assessment was first performed to see which one was best for classificating between the three classes. The best-performing model was then further optimized using the selected dataset combination. A randomized search with cross-validation was applied to tune the main hyperparameters, including the regularization strength C and the penalty type. Regularization was used to control model complexity and improve generalization across heterogeneous datasets, reducing the risk of overfitting. Both L1 and L2 penalties were evaluated, where L1 promotes sparsity by driving some coefficients to zero, while L2 reduces coefficient magnitude while retaining all features. The optimal configuration was selected based on the highest Macro F1-score obtained during cross-validation.

C. Multilabel Classification

1) Experiment 8: All Datasets Combined with Classical ML: This experiment was conducted under the assumption that, given the presence of seven classes, combining all available datasets provides sufficient representation for each class (Appendix A.B Figure 11). This ensures class variability and allows the models to learn all subclasses more effectively under the inherent class imbalance observed in individual datasets.

2) Experiment 9: ML on Teknon with Clinical Variables: The same machine learning methodology was applied using only the Teknon dataset, including its clinical variables as additional input features. The objective was to evaluate whether clinical information from Teknon improves performance in the multilabel classification setting. An additional hyperparameter optimization was conducted for XGBoost in order to assess whether tuning the model further improves performance. XGboost was chosen to be optimized because os its flexibility when in comes to hyperparameter searching.

D. Unsupervised Learning

An unsupervised learning framework was applied across all datasets (Teknon, Clinic, China, and Simulated) to explore the intrinsic structure and separability of the data. This analysis was performed independently of the three downstream classification tasks, with the objective of assessing whether the datasets exhibit an inherent organization consistent with the supervised learning formulations.

Data Preprocessing All datasets were reshaped to a 2D format when required and standardized using z-score normalization to ensure feature comparability across sources.

Dimensionality Reduction Principal Component Analysis (PCA) was applied to obtain a 10-dimensional latent representation used for clustering, preserving variance while reducing noise. A 2D PCA projection was additionally used for visualization purposes.

Clustering K-Means clustering was performed in the latent space. The number of clusters was set equal to the number of ground-truth classes (LVOT/RVOT or extended labels depending on the task definition) solely for evaluation purposes.

Cluster-to-Class Alignment Due to label permutation invariance, the Hungarian algorithm (linear sum assignment) was used to compute an optimal one-to-one mapping between cluster assignments and true labels based on the confusion matrix. Clustering performance was then evaluated using accuracy after alignment and confusion matrices.

IV. Results

A. Task 1: Binary Classification

Raw Data

1) Results Experiment 1: CNN - Dataset Assessment: The best performance was achieved by combining all datasets, reaching a Macro F1-score of 0.602 (confusion matrix provided in Appendix E.A. Figure 14). Although performance remained modest, results were above chance level. Dataset composition had a limited impact, with only a slight benefit from including external datasets (Appendix E.A, Table VIII).

2) Results Experiment 2: CNN + MLP: Based on the dataset assessment, all datasets were combined (Macro F1-score: 0.602) to increase class variability and improve generalization. Incorporating an MLP with Teknon clinical variables improved performance, increasing the Macro F1-score to 0.72 (Appendix E.B. Table IX).

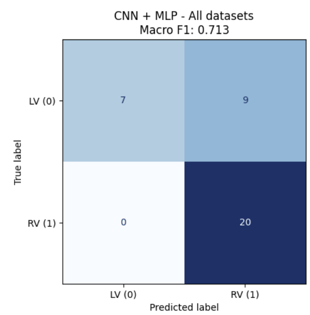

Fig. 2: Confusion Matrix for CNN + MLP.

The SHAP analysis of the MLP branch (RV class) using only clinical and demographic features shows that *PVC_transition* is the most influential variable. Height and weight also rank highly, suggesting a potential association with body habitus. *CLINICAL_SCORE* and age follow in importance, while binary comorbidities (HTA, DM, DLP, sleep apnea, smoking, COPD) contribute minimally, with SHAP values concentrated near zero. Overall, only *PVC_transition* shows occasional high-magnitude effects (Appendix E.C. Figure 15).

3) Results Experiment 3: Classical ML - Dataset Assessment: Logistic Regression achieved the best overall performance, with the highest Macro F1-score obtained using the Simulated and Teknon datasets (Configuration 1), reaching 0.621 (More details provided in Appendix E.D. Table X).

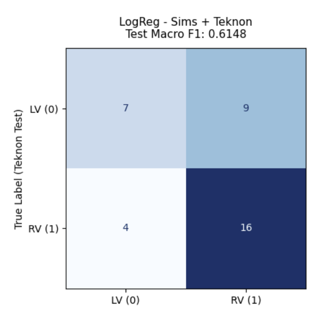

Fig. 3: Confusion Matrix for Teknon + Sims combination using ML.

Feature Based Approach

4) Results Experiment 4: ML - Data Assessment: Feature-based representations improved performance across all models compared to the raw data approach. The highest Macro F1-score was achieved by XGBoost using the China + Teknon combination, reaching 0.74 (Appendix E.E (Table XI). Using this best-performing configuration, the optimized XGBoost model achieved a Macro F1-score of 0.7926.

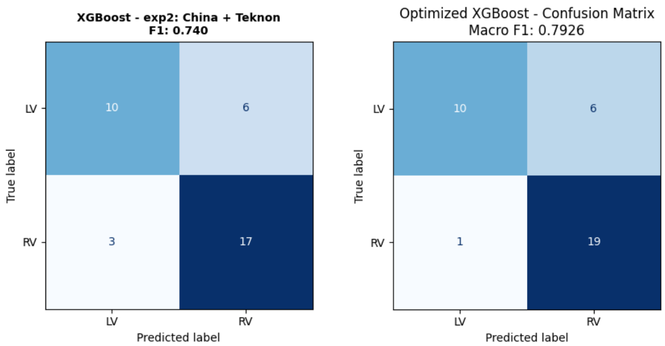

Fig. 4: XGBoost performance before and after hyperparameter optimization. China + Teknon dataset with Clinical Variables.

SHAP analysis on the combined Teknon–China dataset (binary classification) shows that *PVC_transition* is the dominant feature, exhibiting the largest SHAP magnitude and clearly separating the classes. The remaining variables (e.g., *I_RS_ratio*, *V5_RMS*, *I_RMS*) contribute marginally, with values tightly centered around zero, indicating limited discriminative power. Overall, the model is primarily driven by *PVC_transition*, with only minor contributions from the rest of the feature set (Appendix E.F. Figure 16).

5) Results Experiment 5: ML on best data combination with clinical variables from Teknon: The results indicate a clear decrease in performance after applying missing value imputation. Among the evaluated models, Logistic Regression achieved the highest test performance with a Macro F1-score of 0.6099 (Appendix E.G Table XII and Appendix E.G Figure 17). All models exhibited lower performance compared to previous experiments, suggesting that the imputation process negatively affected the predictive power of the clinical variables.

6) Results Experiment 6: ML on Teknon Dataset with Clinical Variables: To further evaluate the impact of clinical variables, the experiment was extended using the Teknon dataset enriched with clinical features. Clinical variables slightly improved the performance of XGBoost and Random Forest, while SVM and Logistic Regression showed marginal decreases. Overall, XGBoost remained the best-performing model.

Model	Macro F1	+ Clinical Macro F1
XGBoost	0.663	0.682
Random Forest	0.625	0.641
SVM-RBF	0.556	0.490
Logistic Regression	0.663	0.649

TABLE III: Comparison of Macro F1 scores using Teknon features only vs. Teknon + clinical variables.

After hyperparameter optimization, XGBoost achieved the best results of the study, with a cross-validation Macro F1-score of 0.723 and a test Macro F1-score of 0.785. Test accuracy, precision, and recall reached 0.778, 0.857, and 0.750, respectively.

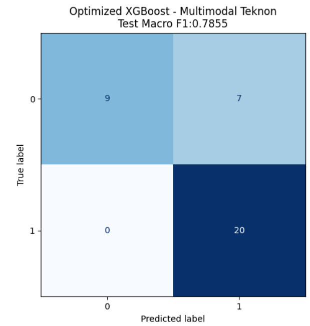

Fig. 5: Confusion Matrix for Teknon with Clinical Variables and Optimized XGBoost.

SHAP analysis for binary classification on the Teknon dataset using ECG and clinical features shows that *PVC_transition* is the dominant predictor, with low values strongly associated with one class and high values with the opposite class, indicating high discriminative power. *aVR_skewness* and age also provide relevant contributions, while clinical variables such as BMI and sex show moderate importance. Overall, ECG morphology features remain the primary source of predictive signal, with clinical variables offering secondary support (Appendix E.H Figure 18).

B. Task 2: Three-label Classification

1) Results Experiment 7: Machine Learning Data Assessment and Model Optimization: A dataset assessment identified the best configuration for the three-label classification task as the combination of all datasets using Logistic Regression, achieving a Macro F1-score of 0.409 (above chance level, 0.33; Table XIII).

After hyperparameter optimization, the Logistic Regression model achieved a test Macro F1-score of 0.407. Although slightly lower than the dataset assessment result, the confusion matrices (Figure 6) show a more balanced prediction distribution, reducing overprediction of RVOT and improving LVOT detection, at the expense of a slight decrease in CC classification performance.

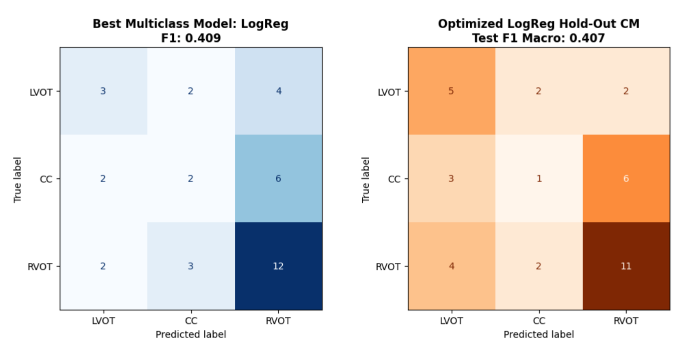

Fig. 6: LogReg performance before and after hyperparameter optimization.

SHAP analysis of the optimized Logistic Regression model for RVOT classification shows that *V2_R_amp* and *V1_R_amp* are the most influential features, with higher values strongly associated with RVOT prediction, consistent with known ECG patterns. *aVL_R_amp* and *V1_skewness* also contribute to the decision boundary. *PVC_transition* appears with lower importance, while *V5_R_amp* shows scattered high-impact outliers that occasionally drive predictions toward RVOT (Appendix E.I Figure 19).

C. Task 3: Multilabel Classification

1) Results Experiment 8: All Datasets Combined with Classical ML: For the multilabel classification task, no dataset assessment was performed due to severe class imbalance in most individual combinations; only the full dataset combination provided a meaningful representation of all subclasses.

All four machine learning models were evaluated on this setting. Logistic Regression achieved the best performance, with a Macro F1-score of 0.181 and was the only model to clearly exceed chance level (0.143), indicating a weak but consistent predictive signal (Appendix E.J Table XIV).

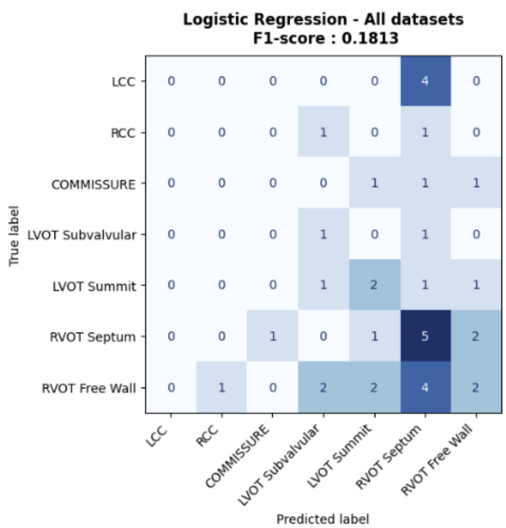

Fig. 7: Confusion Matrix for all the Datasets with Logistic Regression.

Logistic Regression on ECG features only, with SHAP values averaged across 7 classes. The dominant features are amplitude-based: V2_R_amp, V3_R_amp, aVL_R_amp. The SHAP scale is large (up to ±20), which is typical for Logistic Regression (operates in log-odds space). The averaging across classes dilutes class-specific patterns, but precordial R-wave amplitudes clearly matter most globally (Appendix E.J Figure 20).

2) Results Experiment 9: ML on Teknon with Clinical Variables: To evaluate the impact of clinical variables, the Teknon dataset was used with clinical variables and used to train the four machine learning models under the same experimental setup as previous experiments. Logistic Regression achieved the best performance with a Macro F1-score of 0.223. Random Forest showed a slight increase compared to the previous experiment, reaching 0.152, while XGBoost and SVM remained below chance-level performance (Appendix E.K Table XV).

Hyperparameter optimization was applied to XGBoost using the same randomized grid search strategy, but no performance improvement was observed. The best configuration reached a Macro F1-score of 0.159 on the test set (Appendix E.K Figure 19). This indicates that performance is mainly limited by task complexity and data rather than hyperparameter selection.

D. Unsupervised Learning Results

1. Unsupervised Learning for Binary Classification Across all datasets, silhouette scores and clustering agreement metrics indicate limited natural separability in the unsupervised feature space. The TEKNON dataset showed the weakest and most consistent performance (silhouette: 0.1661, matched accuracy: 0.5593), reflecting high variability and noise in real clinical recordings. Overall, these results suggest that the handcrafted feature space does not naturally separate LV and RV origins, limiting the effectiveness of unsupervised clustering.

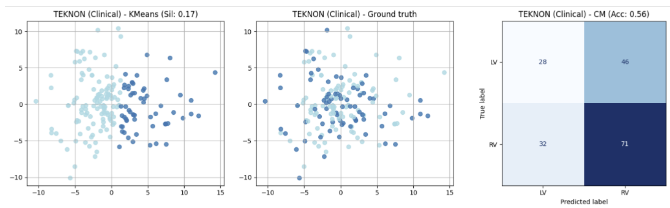

Fig. 8: Unsupervised Learning for Teknon Dataset. Binary Classification.

Dataset	Silhouette Score	Accuracy (Matched)
CARTO (Clinic)	0.4062	0.5116
CHINA (Database)	0.2854	0.8679
SIMS (Synthetic)	0.1902	0.8277
TEKNON (Clinical)	0.1661	0.5593

TABLE IV: Clustering performance across datasets using silhouette score and matched accuracy for Binary Classification.

2. Unsupervised Learning for Multilabel Classification In addition to matched accuracy, the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) were computed to provide class-imbalance-robust evaluations of clustering quality. Results are summarized in Table V. In the multi-class setting, performance further decreases across all datasets. TEKNON again performs worst (accuracy: 0.2486, ARI: 0.0054, NMI: 0.0712), with ARI near zero indicating no meaningful correspondence between clusters and anatomical labels. CARTO shows moderate structure (NMI: 0.3022) but low accuracy, while CHINA is similarly affected by class imbalance.

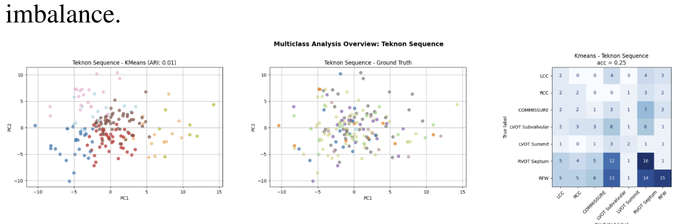

Fig. 9: Unsupervised Learning for Teknon Dataset. Multilabel Classification.

Dataset	Accuracy (Matched)	ARI	NMI
SIMS (Simulations)	0.3582	0.1019	0.2285
CHINA (Database)	0.3003	0.0858	0.2003
CARTO (Clinical Mapping)	0.2093	0.1042	0.3022
Teknon Sequence	0.2486	0.0054	0.0712

TABLE V: Clustering performance across datasets using silhouette score and matched accuracy for Multilabel Classification.

V. Conclusion

A. Binary Classifier

The results show that engineered features provide a stronger predictive signal than raw ECG signals. The 73 morphological and statistical variables capture informative patterns that convolutional filters, limited by sample size, cannot extract effectively.

In the Deep Learning experiments (1 and 2), adding clinical variables via an MLP branch improved the Macro F1-score from 0.602 to 0.72, confirming that demographic and clinical data contribute relevant complementary information. SHAP analysis identifies *PVC_transition* as the dominant feature, which is clinically consistent with its role as the standard criterion for RVOT vs LVOT differentiation.

In classical machine learning, the contribution of clinical variables is less pronounced. On the Teknon cohort (Table III), adding clinical features to the 73 ECG variables yields only a marginal improvement in Macro F1-score (0.663 to 0.682 with XGBoost). This suggests that clinical data plays a secondary, supportive role rather than adding strong independent information. However, SHAP analysis (Figure 18) shows that age ranks highly in importance, with BMI and sex contributing to a lesser extent, while *PVC_transition* remains the main driver.

The optimal data configuration depends on the model type. For feature-based machine learning, combining China + Teknon yields the best performance (Macro F1 = 0.74 with XGBoost), whereas the CNN benefits from the full dataset (CARTO + China + Sims + Teknon). This reflects fundamental differences in learning mechanisms: deep models require large and diverse datasets to learn hierarchical representations from raw signals, while feature-based models rely on stable handcrafted representations and are negatively affected by heterogeneous or synthetic data.

B. Three-Label Classifier

Extending the problem to three classes increases complexity and reduces overall performance. The best results are again obtained using the full dataset, largely due to class distribution effects.

For Logistic Regression, although the overall Macro F1-score is slightly lower (Figure 6), the prediction distribution improves by reducing the strong bias toward RVOT. The lower score is partly explained by Macro F1 equally weighting all classes, meaning errors in minority classes (e.g., Commissure) are heavily penalized. Additionally, the test split (last 20% of the dataset) introduces class imbalance, limiting the fairness of the evaluation.

SHAP analysis (Figure 19) further shows that *PVC_transition*, previously dominant in binary classification, loses importance in this setting. This indicates that global features effective for lateralization are less informative when distinguishing finer anatomical subcategories.

C. Sublocation Classifier

Multilabel classification represents an even more challenging task, with a clear drop in performance. Unsupervised clustering of ground truth labels shows significant overlap, highlighting the intrinsic difficulty of separating fine-grained anatomical locations despite being a simplified representation.

This limitation is also linked to the feature design: the 73 ECG features were optimized for binary lateralization and therefore lack the resolution required for detailed multilabel discrimination.

A methodological shift is also observed: while XGBoost performs best in binary classification, Logistic Regression is superior in multi-class and multilabel settings. This is explained by model complexity: XGBoost's high flexibility leads to overfitting under class imbalance and limited data, whereas Logistic Regression provides stronger regularization, resulting in more stable and evenly distributed predictions across classes.

D. Limitations and Future Improvements

Several limitations should be considered when interpreting the results. First, the models were developed and evaluated primarily on data from Hospital Teknon, with the goal of supporting clinical decision-making in this specific setting. Therefore, external validity to other hospitals, populations, or acquisition protocols is not guaranteed without further validation.

A second limitation is the heterogeneity across datasets. Although combining sources improved performance in some experiments, results were inconsistent due to substantial differences in data structure and annotation. In particular, the Teknon dataset includes patient-level clinical variables and multiple PVC beats per patient, whereas external datasets provide only isolated QRS complexes without clinical context. This mismatch introduces distribution shifts that can limit the benefit of data integration.

Given this context, future work should prioritize expanding a larger and more representative Teknon dataset, including both ECG signals and clinical variables. Since the target application is hospital-specific, strengthening local data quality and volume may be more beneficial than incorporating additional heterogeneous external sources.

Finally, the results indicate that fine-grained site-of-origin classification remains a challenging problem. The limited performance in multilabel settings suggests that current feature representations may not fully capture the subtle electrophysiological differences between ventricular origins. Future studies should focus on identifying more specific morphological and electrophysiological descriptors better suited for this level of anatomical granularity.

References

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Appendix A
Dataset Distribution

A. Distribution of Binary Target per Dataser

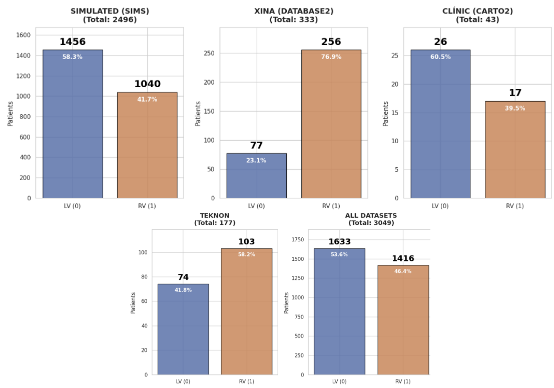

Fig. 10: Distribution of LVOT and RVOT across datasets.

B. Distribution of Multiclass Target per Dataset

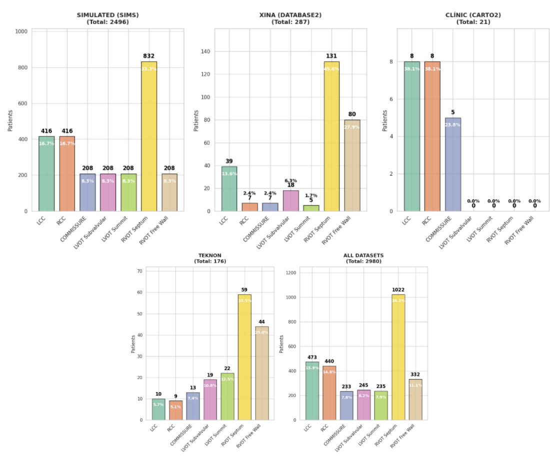

Fig. 11: Multilabel class distributions across the four datasets included in this study.

Appendix B
Clinical Variables Analysis

A. Missing Value Analysis of Clinical Variables from Teknon

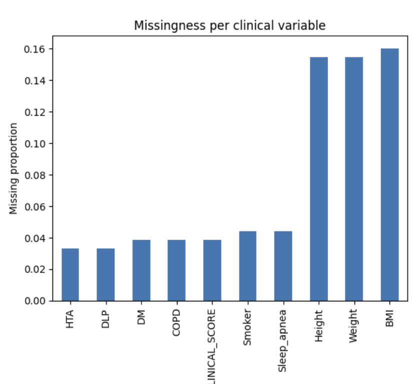

Fig. 12: Distribution of missing values from clinical variables for Teknon dataser.

Appendix C
Methods

A. Convolution Neural Network Architecture

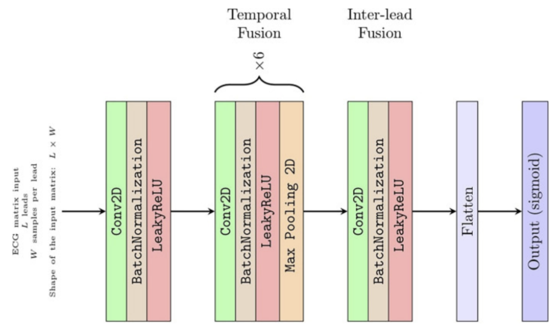

Fig. 13: Convolution Neural Network Architecture from Saglietto et al.

B. Feature Engineering

1) Extracted Features form the QRS Complexes

Feature	Description / Utility
RMS	Signal power; captures overall electrical amplitude magnitude.
Kurtosis	Peakedness of amplitude distribution; high values indicate sharp QRS complexes.
Skewness	Asymmetry of the signal; indicates whether deflections lean positive or negative.
Signal asymmetry	Quantifies waveform directional imbalance across the cycle.
Peak negative deflection	Depth of the S wave in each lead.
Peak positive deflection	Height of the R wave in each lead.
PVC (Premature Ventricular Contraction)	Captures R/S transition pattern and encodes ventricular depolarization direction, providing spatial information to distinguish RVOT/LVOT origins.

TABLE VI: ECG feature definitions and physiological interpretation.

2) PVC Transition Calculation The PVC transition was determined using the R/S ratio across the precordial leads. Specifically, for each lead L_i, the ratio is computed as:

$$\frac{R}{S}(L_i) = \frac{R_{\text{amplitude}}(L_i)}{S_{\text{amplitude}}(L_i)} \tag{1}$$

The transition point is defined as the first precordial lead where:

$$\frac{R}{S}(L_i) > 1 \tag{2}$$

Appendix D
Experiments Summary

A. Experiments Summery

Exp.	Task	Approach	Objective
1	Binary (LVOT/RVOT)	CNN (Raw ECG)	Assess the impact of different dataset combinations using raw ECG signals.
2	Binary (LVOT/RVOT)	CNN + MLP	Train a multimodal model using all datasets and incorporating Teknon clinical variables.
3	Binary (LVOT/RVOT)	Classical ML (Raw ECG)	Evaluate classical machine learning models under the same dataset assessment framework.
4	Binary (LVOT/RVOT)	Feature-Based ML	Perform dataset assessment using 73 extracted ECG features.
5	Binary (LVOT/RVOT)	Feature-Based ML + Clinical Variables	Evaluate the best dataset combination including Teknon clinical variables with missing-value imputation and hyperparameter optimization.
6	Binary (LVOT/RVOT)	Feature-Based ML + Clinical Variables	Assess the predictive value of Teknon clinical variables using only the Teknon dataset.
7	Three-Class Classification	Feature-Based ML	Perform dataset assessment and optimize the best-performing model through hyperparameter tuning.
8	Multilabel Classification	Feature-Based ML	Train and evaluate classical ML models using all datasets combined.
9	Multilabel Classification	Feature-Based ML + Clinical Variables	Evaluate the impact of Teknon clinical variables and perform additional XGBoost optimization.
10	Unsupervised Learning	PCA + K-Means	Explore the intrinsic structure of the datasets through dimensionality reduction and clustering.

TABLE VII: Summary of the experiments conducted in this study.

Appendix E
Graphic Support for Results Section

A. Results Experiment 1: Convolutional Neural Network - Dataset Assessment

Configuration	CNN F1
Config 0	0.580
Config 1	0.562
Config 2	0.357
Config 3	0.496
Config 4	0.535
Config 5	0.446
Config 6	0.474
Config 7	**0.602**

TABLE VIII: CNN F1-score across different dataset configurations.

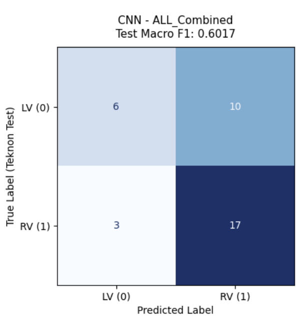

Fig. 14: Confusion Matrix for all the datasets combined configuration.

B. Results Experiment 2: Convolutional Neural Network with Multilayer Perceptron

Class	Precision	Recall	F1-score
LV	1.00	0.44	0.61
RV	0.69	1.00	0.82
Accuracy		0.75	
Macro avg	0.84	0.72	0.71
Weighted avg	0.83	0.75	0.72

TABLE IX: Final test classification results using CNN + MLP on all datasets combined.

C. Shap Analysis for CNN + MLP

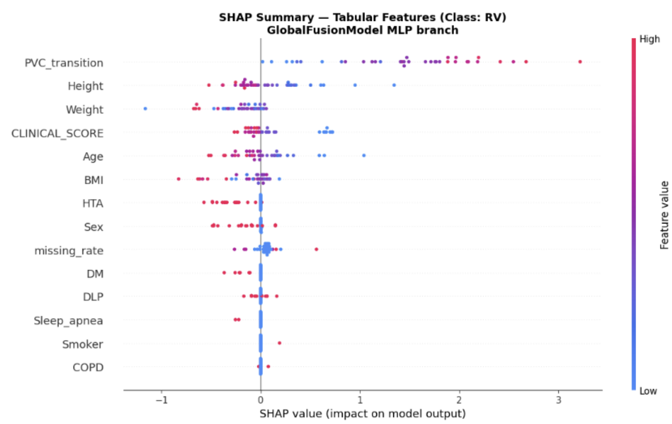

Fig. 15: Shap Analysis for CNN + MLP.

D. Results Experiment 3: Classical Machine Learning - Dataset Assessment

Config	RF (Macro F1)	LogReg (Macro F1)	SVM-RBF (Macro F1)	XGBoost (Macro F1)
Config 0	0.551	0.556	0.428	0.538
Config 1	0.508	**0.621**	0.507	0.514
Config 2	0.447	0.563	0.400	0.479
Config 3	**0.559**	0.599	0.445	0.486
Config 4	0.496	0.532	0.435	0.531
Config 5	0.538	0.617	**0.519**	**0.544**
Config 6	0.511	0.543	0.412	0.488
Config 7	0.515	0.553	0.427	0.533

TABLE X: Macro F1-score comparison across machine learning models and dataset configurations.

E. Results Experiment 4: Machine Learning - Dataset Assessment

Configuration	XGBoost	RandomForest	SVM-RBF	LogReg
Config 0	0.649	0.607	0.542	0.663
Config 1	0.682	0.630	0.542	0.486
Config 2	**0.740**	**0.713**	0.585	**0.688**
Config 3	0.733	0.607	**0.602**	0.649
Config 4	0.674	0.630	0.542	0.591
Config 5	0.649	0.585	0.540	0.486
Config 6	0.657	0.649	0.585	**0.688**
Config 7	0.708	0.673	0.542	0.452

TABLE XI: Macro F1-score comparison across machine learning models and dataset configurations after feature extraction.

F. Shap Analysis for China + Teknon with Optimized XGBoost

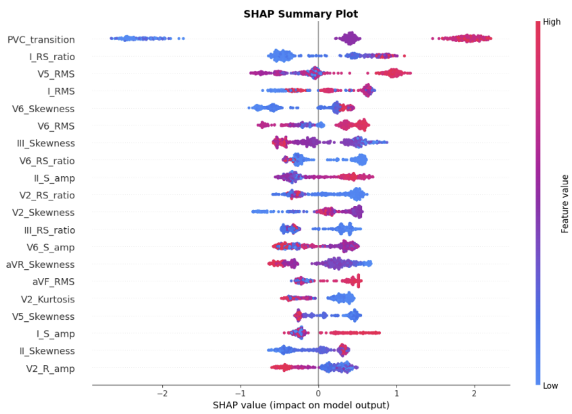

Fig. 16: Shap Analysis for Optimized XGBoost. Best Combination of datasets: China + Teknon.

G. Results Experiment 5: Machine Learning on Best Data Combination with Clinical Variables from Teknon

Model	Macro F1-score
XGBoost	0.5670
Random Forest	0.5154
Support Vector Machine	0.4375
Logistic Regression	**0.6099**

TABLE XII: Performance on the combined China and Teknon dataset using Teknon clinical variables after missing value imputation.

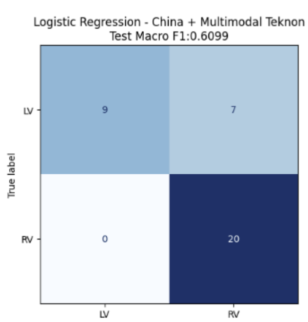

Fig. 17: Confusion Matrix for Logistic Regression for combined China and Teknon dataset using Teknon clinical variables

H. Results Experiment 6: Machine Learning on Teknon Dataset with Clinical Variables

Shap Analysis for Clinical Varibales from Teknon + Clinical Variables

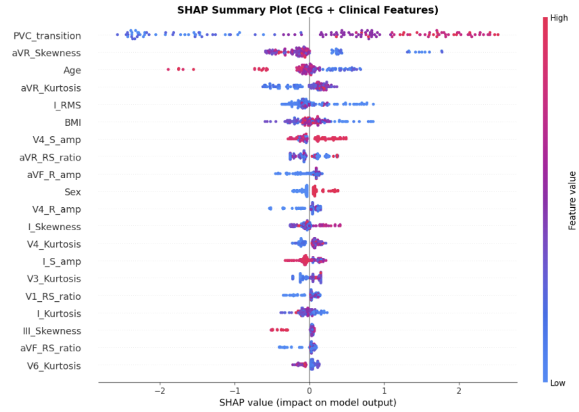

Fig. 18: Shap Analysis for Teknon dataset with Teknon's Clinical Variables.

I. Results Experiment 7: Machine Learning Data Assessment and Model Optimization

Experiment	XGBoost	RandomForest	SVM-RBF	LogReg
Config 0	0.302	0.209	0.214	0.383
Config 1	0.365	0.214	0.214	0.374
Config 2	0.394	0.283	0.285	0.350
Config 3	0.320	0.218	0.214	0.330
Config 4	0.269	0.205	0.214	0.392
Config 5	0.304	0.214	0.214	0.373
Config 6	0.274	0.218	0.285	0.340
Config 7	0.342	0.214	0.214	**0.409**

TABLE XIII: Macro F1-score for multiclass classification (3 classes) using feature-based ML models on Teknon and external datasets.

Shap Analysis for three-class classification for Logistic Regression on all Datasets Combined

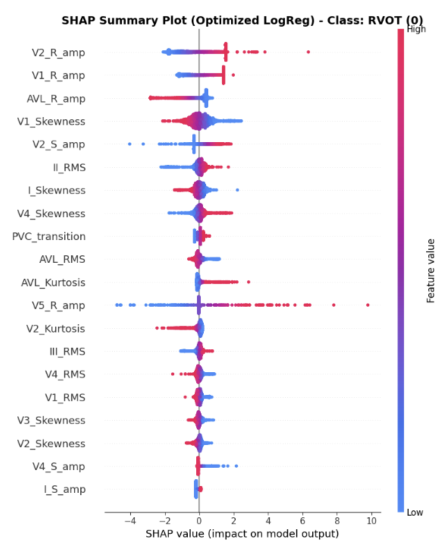

Fig. 19: Shap Analysis for Three-Classes Classification.

J. Results Experiment 8: All Datasets Combined with Classical Machine Learning

Model	Test Macro F1
Logistic Regression	**0.181**
XGBoost	0.111
Random Forest	0.082
Support Vector Machine	0.082

TABLE XIV: Final test-set Macro F1 for multiclass classification (feature-based ML, all datasets).

Shap Analysis for All datasets Combined Logistic Regression Model

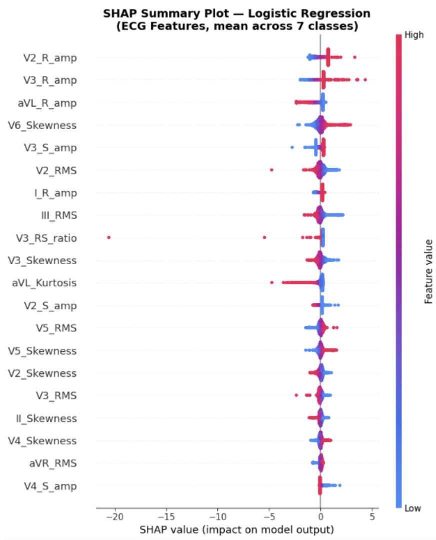

Fig. 20: Shap Analysis for Optimized Logistic Regression for Multilabel Classification.

K. Results Experiment 9: Machine Learning on Teknon with Clinical Variables

Model	Macro F1
XGBoost	0.089
Logistic Regression	0.223
Random Forest	0.152
SVM	0.029

TABLE XV: Macro F1-score for multiclass classification using only Teknon clinical variables (feature-based approach).

Confussion Matrix for Multilabel Classification for Teknon with Clinical Variables and OPtimized XGBoost.

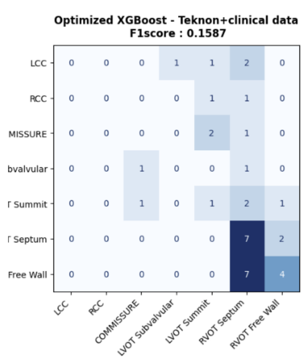

Fig. 21: Confusion Matrix for Teknon with Clinical Variables and Optimized XGBoost.

Shap Analysis for Multilabel Classification for Teknon with Clinical Variables and OPtimized XGBoost.

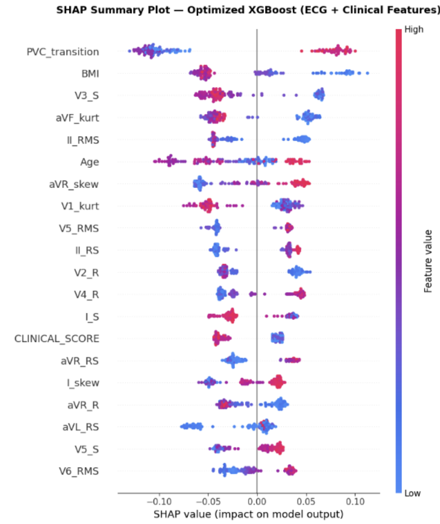

Fig. 22: Shap Analysis for Teknon dataset with clinical variables for Multilabel Classification.

Appendix F
Git-Hub Repository

All the ipynb and sources are available in:
https://github.com/Maria65432/CompBioMed_2026